Detection statistic n_σ as a function of chirp mass and SFT duration in the remapped $(t, f^{-8/3})$ search

We derive the dependence of the detection statistic n_σ on the chirp mass $\mathcal{M}$ and on the SFT duration $T_{\rm SFT}$, following the steps of Ref. [arXiv:2401.02314] and indicating where the present search differs from it.

1 Definition of n_σ

The Viterbi algorithm accumulates the normalised power of the pixels along the recovered track. In the absence of a signal, the sum over $N_{\rm SFT}$ pixels has a mean $\mu \propto N_{\rm SFT}$ and a standard deviation $\sigma \propto \sqrt{N_{\rm SFT}}$. A signal increases the sum by its total optimal squared signal-to-noise ratio $\rho^2_{\rm tot}$, so that

$$n_\sigma = \frac{\rho^2_{\rm tot}}{\sigma} \propto \frac{\rho^2_{\rm tot}}{\sqrt{N_{\rm SFT}}}. \tag{1}$$

This is Eq. (2.21) of Ref. [arXiv:2401.02314]. The noise background measured in the present search follows $\sigma \propto T_{\rm SFT}^{-0.503}$, i.e. $\sigma \propto \sqrt{N_{\rm SFT}}$.

2 Signal-to-noise ratio of the signal

For an inspiral, $|\tilde{h}(f)|^2 \propto \mathcal{M}^{5/3} f^{-7/3}$, and therefore

$$\rho^2_{\rm tot} \propto \mathcal{M}^{5/3} \int \frac{df}{f^{7/3} S_n(f)}, \tag{2}$$

where the integral extends over the frequencies at which the Viterbi algorithm is able to follow the signal. This step is identical to Ref. [arXiv:2401.02314].

3 Number of noise steps

In Ref. [arXiv:2401.02314] only the time during which the signal is present is considered, $N_{\rm SFT} = T_{\rm CBC}/T_{\rm SFT}$. In the present search, the Viterbi algorithm runs over the whole data chunk, so that $N_{\rm SFT} = T_{\rm obs}/T_{\rm SFT}$ with $T_{\rm obs} = 32768$ s fixed. Since $T_{\rm obs}$ depends neither on the chirp mass nor on $T_{\rm SFT}$, it enters only as a constant:

$$n_\sigma \propto \sqrt{\frac{T_{\rm SFT}}{T_{\rm obs}}}\, \mathcal{M}^{5/3} \int \frac{df}{f^{7/3} S_n(f)}. \tag{3}$$

In both cases the factor $\sqrt{T_{\rm SFT}}$ appears: longer SFTs imply fewer noise steps.

4 Frequency range over which the signal is tracked

This is the key difference between the two approaches.

Linear frequency map (Ref. [arXiv:2401.02314]). The signal is tracked as long as it remains within a single linear frequency bin during one SFT, i.e. up to the frequency $f_*(T_{\rm SFT})$ defined by $\dot{f}(f_*)\,T_{\rm SFT}^2 = 1$; above it, the signal is lost:

$$n_\sigma^{\rm lin} \propto \sqrt{T_{\rm SFT}} \int_{f_0}^{f_*(T_{\rm SFT})} \frac{df}{f^{7/3} S_n(f)}. \tag{4}$$

Two effects compete: increasing $T_{\rm SFT}$ increases $\sqrt{T_{\rm SFT}}$ but lowers f_*, removing high frequencies from the integral. The optimum lies where both effects balance, which yields Eq. (14) of the paper.

Remapped map (present search). In the $(t, f^{-8/3})$ map the signal is a straight line with constant drift. The Viterbi algorithm therefore follows it either over the *whole* band or not at all, depending on whether the signal moves by at most one remapped bin per step, i.e. on whether $T_{\rm SFT} \le T_{\rm SFT}^{\rm lim}$. The integral no longer depends on $T_{\rm SFT}$, and only a cutoff remains:

$$n_\sigma \propto \sqrt{T_{\rm SFT}} \int_{f_0}^{f_1} \frac{df}{f^{7/3} S_n(f)}\, \Theta\Big(T_{\rm SFT}^{\rm lim} - T_{\rm SFT}\Big). \tag{5}$$

5 Final expression

Normalising to the maximum over $T_{\rm SFT}$ at each chirp mass, as in Fig. 2 of Ref. [arXiv:2401.02314], the integral and all constant factors cancel:

$$\frac{n_\sigma(\mathcal{M}, T_{\rm SFT})}{\max_{T_{\rm SFT}} n_\sigma} = \begin{cases} \sqrt{T_{\rm SFT}/T_{\rm SFT}^{\rm lim}(\mathcal{M})}, & T_{\rm SFT} \le T_{\rm SFT}^{\rm lim}(\mathcal{M}), \\ 0, & T_{\rm SFT} > T_{\rm SFT}^{\rm lim}(\mathcal{M}), \end{cases} \tag{6}$$

with

$$T_{\rm SFT}^{\rm lim} \simeq 17.7\ {\rm s} \left(\frac{\mathcal{M}}{10^{-2}\,M_\odot}\right)^{-5/6} \qquad (f_0 = 61.1\ {\rm Hz},\ f_1 = 126.8\ {\rm Hz}). \tag{7}$$

6 Validation with injections

Equation (6) was tested with 75 injections into O3b data: five chirp masses (5.6×10^{-4}, 1.6×10^{-3}, 3.2×10^{-3}, 1.8×10^{-2} and $5.0\times10^{-2}\,M_\odot$), each at five distances ranging from loud signals to the detection limit, and three random data chunks per point. For each injection, n_σ was computed for the 13 SFT durations of the search. Since only these discrete values are available, the maximum is reached at $T_{\rm ref}$, the largest $T_{\rm SFT}$ of the grid not exceeding $T_{\rm SFT}^{\rm lim}$, and the prediction becomes $\sqrt{T_{\rm SFT}/T_{\rm ref}}$.

Optimum. For every detected signal the maximum of n_σ is reached at $T_{\rm ref}$, and n_σ drops to the noise level for $T_{\rm SFT} > T_{\rm SFT}^{\rm lim}$, as predicted.

Shape near the optimum. The relevant range is $T_{\rm SFT}$ close to $T_{\rm SFT}^{\rm lim}$, since the search always uses the largest available $T_{\rm SFT} \le T_{\rm SFT}^{\rm lim}$ and consecutive values of the grid differ by a factor of 1.3–1.5. Within $T_{\rm ref}/3 \le T_{\rm SFT} \le T_{\rm ref}$ and for signals with $\max n_\sigma \gtrsim 10$, the measured normalised n_σ differs from $\sqrt{T_{\rm SFT}/T_{\rm ref}}$ by at most 0.13 (in absolute value of the normalised quantity) and by less than 0.09 on average. Equation (6) is therefore a very good approximation in the regime in which the search operates.

Shape far from the optimum. For $T_{\rm SFT} \ll T_{\rm SFT}^{\rm lim}$ the measured n_σ falls below $\sqrt{T_{\rm SFT}}$, increasingly so for weaker signals: with short SFTs the signal-to-noise ratio per step, $\propto \sqrt{T_{\rm SFT}}$, is low and the Viterbi algorithm does not follow the signal in every step. These SFT durations are never selected for the corresponding chirp masses, so this deviation does not affect the choice of $T_{\rm SFT}$.

Dense set of SFT durations. To test the expression close to $T_{\rm SFT}^{\rm lim}$, we built the smallest set of SFT durations for which Eq. (6) guarantees $n_\sigma \geq 0.95 \max n_\sigma$ for every chirp mass in $[2 \times 10^{-4}, 0.1]\, M_\odot$: 45 values from 2.6 to 422 s, consecutive values differing by a factor $\simeq 1.11$. The noise background was measured for every value, and three chirp masses were injected at five distances each, with three data chunks per point. Now $T_{\rm ref}$ lies within 11% of $T_{\rm SFT}^{\rm lim}$ (Fig. 1). For all detected signals the maximum is reached at $T_{\rm ref}$, and n_σ drops to the noise level above $T_{\rm SFT}^{\rm lim}$. Within $T_{\rm ref}/3 \leq T_{\rm SFT} \leq T_{\rm ref}$:

- for $\mathcal{M} = 5.0 \times 10^{-2}\, M_\odot$, the measured curves follow $\sqrt{T_{\rm SFT}/T_{\rm ref}}$ within 0.09;
- for $\mathcal{M} = 3.2 \times 10^{-3}\, M_\odot$, within 0.17, with a mean deviation below 0.04;
- for $\mathcal{M} = 5.6 \times 10^{-4}\, M_\odot$, the measured n_σ lies *above* $\sqrt{T_{\rm SFT}/T_{\rm ref}}$, by 0.10–0.15 on average: the gain flattens close to $T_{\rm SFT}^{\rm lim}$.

Equation (6) is therefore accurate for heavy and intermediate chirp masses, and a conservative lower bound for light ones. One step below $T_{\rm ref}$ in the set, the measured normalised n_σ is 0.93–0.99 (median 0.96), consistent within the scatter between data chunks with the predicted 0.94–0.96.

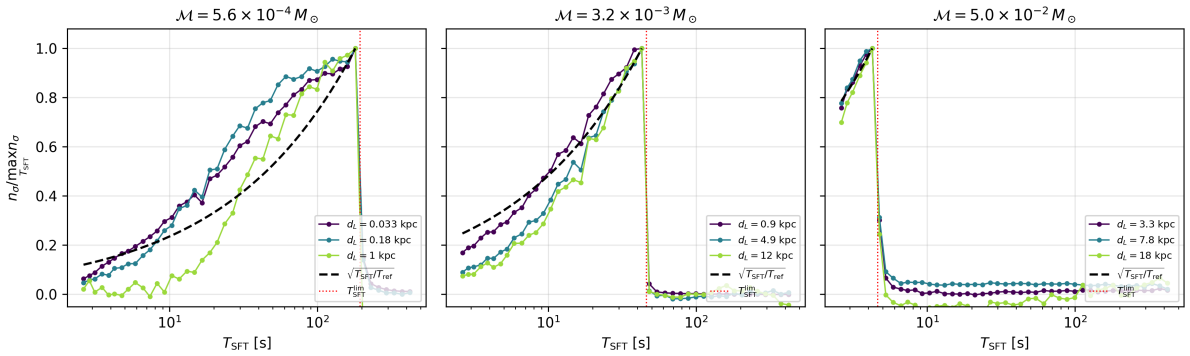


Figure 1: Normalised n_σ as a function of $T_{\rm SFT}$ for the dense set of 45 SFT durations, for three chirp masses and several luminosity distances d_L (median over three data chunks; only detected signals are shown). The dashed line is $\sqrt{T_{\rm SFT}/T_{\rm ref}}$, and the dotted line marks $T_{\rm SFT}^{\rm lim}$.

Table 1 summarises the differences between both approaches.

	Ref. [arXiv:2401.02314] (linear frequency map)	Present search (remapped map)
Noise steps	$T_{\mathrm{CBC}}/T_{\mathrm{SFT}}$	$T_{\mathrm{obs}}/T_{\mathrm{SFT}}$
Tracking limit	$f_*(T_{\mathrm{SFT}})$: one linear bin per SFT	$T_{\mathrm{SFT}}^{\mathrm{lim}}$: one remapped bin per step
Effect of increasing T_{SFT}	Gradual loss of high frequencies	No loss up to $T_{\mathrm{SFT}}^{\mathrm{lim}}$, total loss beyond
Optimum	Balance between $\sqrt{T_{\mathrm{SFT}}}$ and the integral [Eq. (14)]	Largest $T_{\mathrm{SFT}} \leq T_{\mathrm{SFT}}^{\mathrm{lim}}$
Normalised n_σ	Smooth around the optimum	$\sqrt{T_{\mathrm{SFT}}/T_{\mathrm{SFT}}^{\mathrm{lim}}}$ with a sharp cutoff

Table 1: Comparison of the dependence of n_σ on T_{SFT} in Ref. [arXiv:2401.02314] and in the present search.